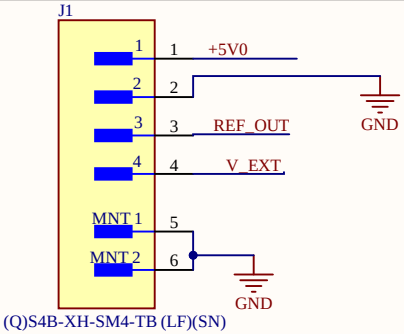
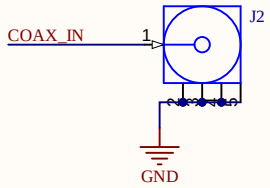
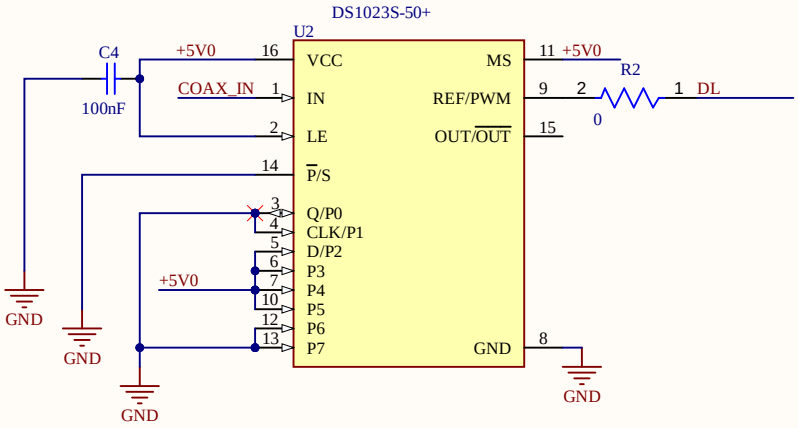
Power Regulation & External Interface

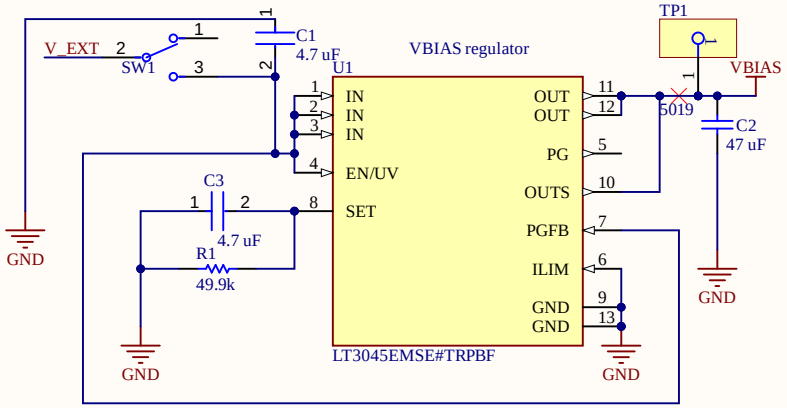




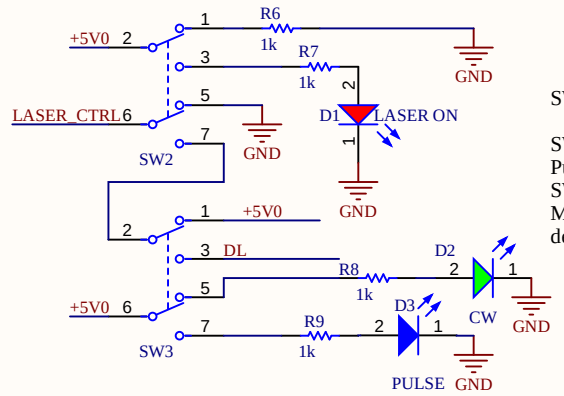


PULSE GENERATOR (DS1023)
- Configuration: One-shot (monostable) pulse generator
- Input: Trigger signal from MCU
- Input: Pulse-width configuration from MCU
- Output: Nanosecond electrical pulse
- Function: Generates a single, precisely timed pulse upon receiving an MCU trigger, with programmable pulse width.

Can be Programmed to have the pulse widths be adjustable over GPIO (Right now set programming) P0-P7 current programming is 30ns



VBIAS REGULATOR (LT3045)
- Input: ACDC adapter supply
- Output: Regulated VBIAS supply
- Function provide laser vias voltage for driving different laser diodes

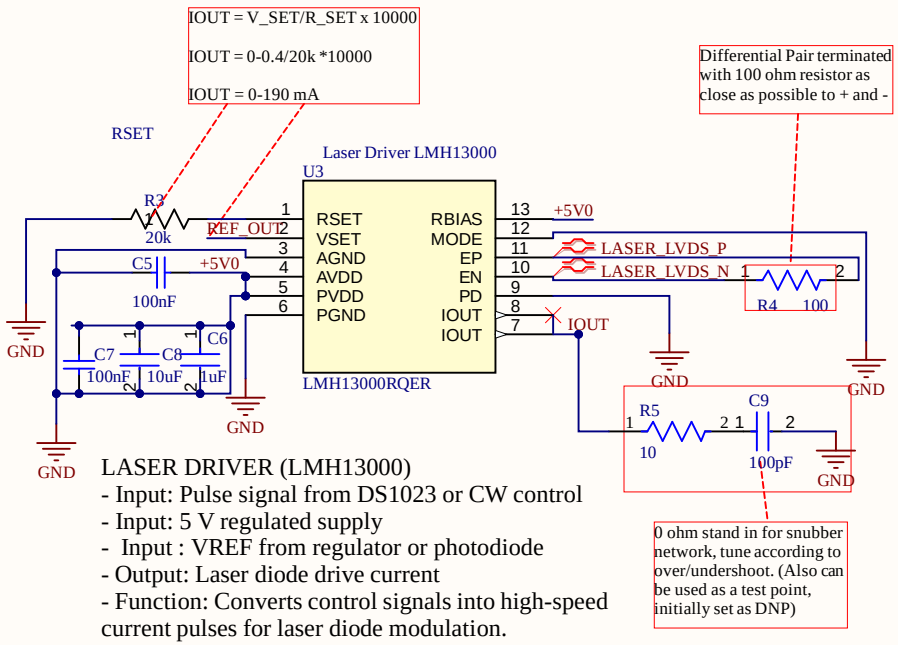


SW1: Enable/Disable

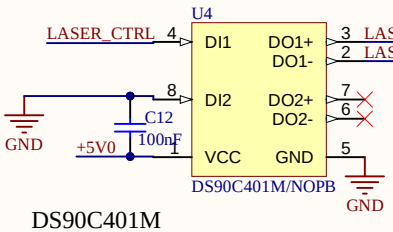
SW2: Switching between Pulsed and CW
SW3: Switching between MCU defined and physically defined pulses

Program: **Quantum Key Distribution**	Drawn By: **Adam Szautner**	Reviewed By: **=RvwdBy**	Revision: *****
Project Title: **Laser Driver**	Project ID: **QKD-LD-002**	Last Modified: **Nov. 1st, 2026**	SHEET 1 OF 2

Laser Driver & Simulations, Differential Pair Splitting



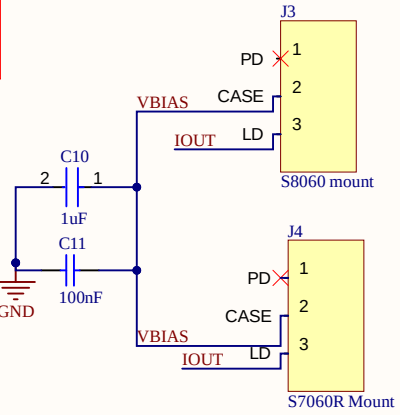
LASER DRIVER (LMH13000)
- Input: Pulse signal from DS1023 or CW control
- Input: 5 V regulated supply
- Input : VREF from regulator or photodiode
- Output: Laser diode drive current
- Function: Converts control signals into high-speed current pulses for laser diode modulation.



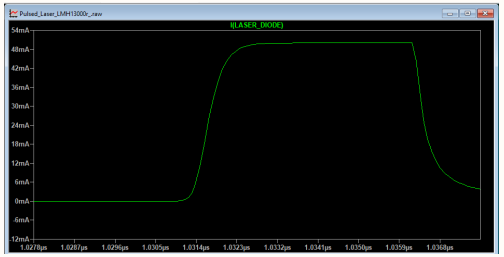
DS90C401M
- Input: pulse or CW input
- Output: Differential pair
- Function: Take pulse input and convert it into a usable signal by the laser driver

VBIAS Cannot open file C:\Users\szaut\Download IOUT

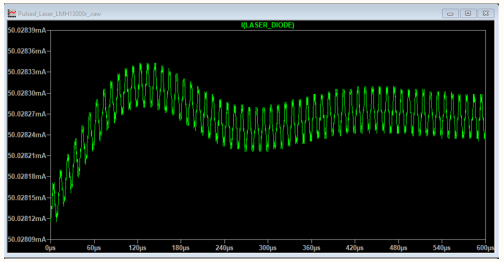
Orientation of the laser diode



Results are based on the LMH13000 SPICE model and a representative laser diode equivalent circuit. Experimental verification is required to confirm pulse fidelity, current stability, and switching performance.



SIM-02 - Simulated laser diode current under 5 ns input pulse modulation. The LMH13000 achieves approximately 50 mA peak current with rapid switching transitions, minimal overshoot, and a small residual falling-edge tail.



SIM-03 - Current Stability Transient simulation of the laser diode current during CW operation. The output stabilizes near 50 mA, with small residual fluctuations following the initial settling period. (nA scale fluctation)

Program: **Quantum Key Distribution**	Drawn By: **Adam Szautner**	Reviewed By: **=RvwdBy**	Revision: **1.0**
Project Title: **Laser Driver**	Project ID: **QKD-LD-002**	Last Modified: **Nov. 1st, 2026**	SHEET 2 OF 2

Board Stack Report